[Total No. of Questions :3] **[2511]-1303101** **Seat No.:**

Society for Computer Technology and Research's

PUNE INSTITUTE OF COMPUTER TECHNOLOGY

IN-SEM EXAMINATION OCT-2025

(S.Y.) (B.Tech.) (CE) (SEM-III)

(1303101) (Data Structures)

(2024 PATTERN)

[Time : 1 Hours] **[Max Marks : 20]**

Instructions to the candidates:

1. *Assume suitable data wherever required.*
2. *Neat diagrams must be drawn wherever necessary.*
3. *Use of non-programmable calculators is allowed.*
4. *[Marks], [Course Outcome] and [Bloom's Taxonomy Level] given at the end of each questions.*
5. ***Attempt Any Two Questions***

Q.1)A) Define Time and Space complexity of algorithm in terms of Big O, Omega and Theta Notation.

```
findFactorial(number):

// INPUT

 // number = A positive integer

// OUTPUT

 // Returns the factorial of the number

 if number = 1: return 1

else: return number * findFactorial(number - 1)

end findFactorial
```

Derive the time and space complexity of the above algorithm. [05] [CO1][L3]

B) Write the pseudocode of sentinel search. Analyze the time and space complexity using frequency count. Find the number of comparisons to find the element 50 in the list 20,30, 40, 15, 50, 60, 10, 20 using sentinel search. [05] [CO1][L4]

Q.2)A) Write the pseudocode for the sorting method of O(nlogn) time complexity for all cases. Give the output of each iterations for the given

list to be sorted using the same algorithm. 10, 20, 15, 30, 5, 48, 9, 8. [06] [CO1][L3]

B) What is an associative array? How to find the element in an associative array? Give with an example. [04] [CO1][L2]

Q.3)A) Write the pseudocode for quick sort and apply and show the output of each iteration on elements 10, 60, 50, 30 20. [04] [CO1] [L3]

B) Compute the iterations of Binary search for the elements 20, 30, 40, 50, 70 and searching element 20. Derive its time complexity. [04] [CO1] [L3]

C) What is collision and overflow in Hashing. When do collision and overflow occur together? [02] [CO1] [L2]

····· End of question paper·····